

Frage 1
Teilweise richtig
Erreichte Punkte
4,50 von 6,00
Frage markieren

Which statements about Factor Analysis (FA) are correct?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. The factors are unique only up to orthogonal transformations (rotations).
- ☒ b. Correlations between observations can only be explained by factors since the noise covariance matrix Ψ is diagonal. ✓
- ☒ c. FA describes the variability of observations in terms of latent variables and noise. ✓
- ☐ d. The factor loading matrix $\mathbf{U}$ describes the independent part (of the covariance of the observations).
- ☒ e. FA decomposes the covariance matrix of the data as $\mathbf{C} = \mathbf{U}\mathbf{U}^T + \Psi$. ✓
- ☐ f. FA assumes Gaussian noise and non-Gaussian latent variables.
- ☐ g. Factors explain the variance of the observations, while the correlation is explained by noise.

Die Antwort ist teilweise richtig.

Sie haben 3 richtig ausgewählt.

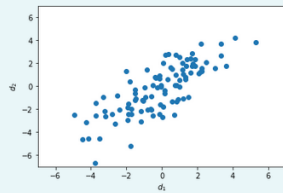
Die richtigen Antworten sind: FA describes the variability of observations in terms of latent variables and noise., FA decomposes the covariance matrix of the data as $\mathbf{C} = \mathbf{U}\mathbf{U}^T + \Psi$, Correlations between observations can only be explained by factors since the noise covariance matrix Ψ is diagonal.

The factors are unique only up to orthogonal transformations (rotations).

Frage 2
Falsch
Erreichte Punkte 0,00 von 6,00
Frage markieren

You are given n data points $\mathbf{x}_1, \dots, \mathbf{x}_n$ with $\mathbf{x}_i \in \mathbb{R}^2$. This data has been drawn from a multivariate Gaussian with density $p_{\text{true}} = \mathcal{N}(\mathbf{0}, \mathbf{C})$ with $\mathbf{C} = \begin{pmatrix} 5.5 & 4.5 \\ 4.5 & 5.5 \end{pmatrix}$.

The data are displayed in the figure below.



The likelihood of the data under the true density evaluates to $\mathcal{L}_{\text{true}}$.

You want to fit a factor analysis model with 1 factor to the data.

But instead of the usual factor analysis assumptions you drop the noise term.

That means your model assumption is that the data has been generated by first sampling $\mathbf{y}_i$ from a standard normal distribution, $\mathbf{y}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The $\mathbf{x}_i$ are generated by $\mathbf{x}_i = \mathbf{U}\mathbf{y}_i$, where $\mathbf{U} \in \mathbb{R}^{2 \times 1}$.

The maximum likelihood of the data for all possible values of $\mathbf{U}$ for this model is $\mathcal{L}_{\text{model}}$. Which of the following are correct:

- ☐ a. $0 \leq \mathcal{L}_{\text{model}} \leq 2\mathcal{L}_{\text{true}}$
- ☐ b. $\sqrt{n}\mathcal{L}_{\text{true}} \leq \mathcal{L}_{\text{model}} \leq n\mathcal{L}_{\text{true}}$
- ☒ c. $0 < \mathcal{L}_{\text{model}} < \mathcal{L}_{\text{true}}$ ✗
- ☐ d. $\frac{1}{n}\mathcal{L}_{\text{true}} \leq \mathcal{L}_{\text{model}} \leq n\mathcal{L}_{\text{true}}$

Your answer is incorrect.

Die richtige Antwort ist: $0 \leq \mathcal{L}_{\text{model}} \leq 2\mathcal{L}_{\text{true}}$

Frage 3

Teilweise
richtigErreichte
Punkte 1,00
von 6,00Frage
markieren

Which statements about projection and scaling methods are correct?

- ☒ a. Non-negative Matrix Factorization (NMF) performs a multiplicative decomposition of a positive semi-definite matrix into two positive semi-definite matrices. ✖
- ☐ b. Locally Linear Embedding (LLE) is a linear embedding method that aims to preserve the neighborhood of each data point.
- ☒ c. Multidimensional Scaling (MDS) aims to find a low-dimensional representation of the data that preserves pairwise Euclidean distances. ✔
- ☐ d. Stochastic Neighbor Embedding (SNE) defines the similarity of two data points as a conditional probability while t-SNE defines it as a joint probability.
- ☒ e. Isomap measures distances between data points as the length of the shortest path along a neighborhood graph. ✔

Your answer is partially correct.

Sie haben 2 richtig ausgewählt.

Die richtigen Antworten sind: Multidimensional Scaling (MDS) aims to find a low-dimensional representation of the data that preserves pairwise Euclidean distances., Isomap measures distances between data points as the length of the shortest path along a neighborhood graph.,

Stochastic Neighbor Embedding (SNE) defines the similarity of two data points as a conditional probability while t-SNE defines it as a joint probability.

Frage 4

Teilweise
richtigErreichte
Punkte 3,00
von 6,00Frage
markieren

Which statements about t-distributed Stochastic Neighborhood Embedding are correct? t-SNE...

- ☒ a. ...is usually trained by gradient descent. ✔
- ☐ b. ...models the neighborhood in both observation and embedding space by a Gaussian distribution.
- ☒ c. ...models the neighborhood of a data point in the observation space proportional to a Student's t-distribution. ✖
- ☒ d. ...minimizes the Kullback-Leibler divergence between distributions in observation space and embedding space. ✔

Your answer is partially correct.

Sie haben zu viele Optionen gewählt.

Die richtigen Antworten sind:

...minimizes the Kullback-Leibler divergence between distributions in observation space and embedding space.,

...is usually trained by gradient descent.

Frage 5

Teilweise
richtigErreichte
Punkte 3,60
von 6,00Frage
markieren

Which statements about Clustering are correct?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. k -means is faster than MoG and not sensitive to initialization.
- ☐ b. Both k -means and MoG assign an observation to a single cluster only.
- ☐ c. The mixing coefficients in a Mixture of Gaussians model are represented by the posteriors.
- ☐ d. The maximum a posteriori estimation is a way to avoid near-zero variances and singular solutions in the maximum likelihood estimation of mixture models.
- ☐ e. Given an observation $\mathbf{x}$, a component j and a parameter vector θ , the posterior $p(j | \mathbf{x}, \theta)$ describes how likely it is that $\mathbf{x}$ was produced by j .
- ☒ f. The EM algorithm for MoG tries to maximize the likelihood with respect to the cluster centers, the covariance matrices and the mixing coefficients. ✔
- ☒ g. The k -means algorithm is a simplification of the Mixture of Gaussians model. ✔
- ☒ h. Clustering is an unsupervised technique that aims at finding separate regions of high data density called clusters. ✔

Die Antwort ist teilweise richtig.

Sie haben 3 richtig ausgewählt.

Die richtigen Antworten sind: Clustering is an unsupervised technique that aims at finding separate regions of high data density called clusters., Given an observation $\mathbf{x}$, a component j and a parameter vector θ , the posterior $p(j | \mathbf{x}, \theta)$ describes how likely it is that $\mathbf{x}$ was produced by j ., The k -means algorithm is a simplification of the Mixture of Gaussians model.,

The EM algorithm for MoG tries to maximize the likelihood with respect to the cluster centers, the covariance matrices and the mixing coefficients.,

The maximum a posteriori estimation is a way to avoid near-zero variances and singular solutions in the maximum likelihood estimation of mixture models.

Frage 6

Falsch

Erreichte Punkte 0,00 von 7,00

Frage markieren

You are given 8 datapoints in $\mathbb{R}^2$,

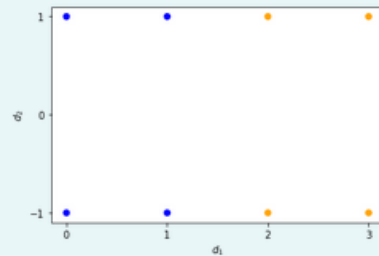
$\{(0, 1), (1, 1), (2, 1), (3, 1), (0, -1), (1, -1), (2, -1), (3, -1)\}$.

Agglomerative hierarchical clustering analysis was done on this data until only two clusters remained with different linkage functions, namely

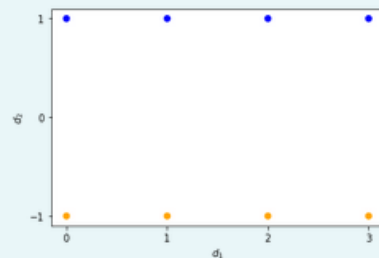
- single linkage: $d_{\min}(A, B) = \min_{a \in A, b \in B} \|a - b\|$
- complete linkage: $d_{\max}(A, B) = \max_{a \in A, b \in B} \|a - b\|$

One of the clusterings below has been generated using single linkage and one using complete linkage.

Clustering 1:



Clustering 2:



Check the correct statements:

- ☒ a. Clustering 1 has been obtained by single linkage ✖
- ☐ b. Clustering 2 has been obtained by single linkage
- ☒ c. Clustering 2 has been obtained using complete linkage ✖
- ☐ d. Clustering 1 has been obtained using complete linkage
- ☒ e. One of the clusterings could have theoretically been the result when using either of the two linkage functions. ✔

Your answer is incorrect.

Die richtigen Antworten sind:

Clustering 1 has been obtained using complete linkage.

Clustering 2 has been obtained by single linkage.

One of the clusterings could have theoretically been the result when using either of the two linkage functions.

Frage 7

Falsch

Erreichte Punkte 0,00 von 7,00

Frage markieren

The data generation in a mixture of Gaussians model is the following:

First one of the Gaussians is chosen according to its weight in the mixture model. Then a sample from the according Gaussian is drawn.

You have fitted a Mixture of two Gaussians to 1D data which resulted in the model

$$p(x) = \alpha_1 f(x; \mu_1, \sigma_1) + \alpha_2 f(x; \mu_2, \sigma_2)$$

where $f(x; \mu, \sigma)$ is the density of a normal distribution with mean μ and standard deviation σ .

To assign a sample to a cluster (one of the Gaussians) we need to calculate the probability that it was "produced" by it. The obtained parameters are:

$$\alpha_1 = 0.3, \alpha_2 = 0.7, \mu_1 = 0, \sigma_1 = 1, \mu_2 = 3, \sigma_2 = 1$$

Some relevant densities are

$$f(1; 0, 1) = 0.242, f(1; 3, 1) = 0.054$$

Calculate the probability that the datapoint $x = 1$ has been produced by the first Gaussian (up to two decimal places).

Antwort: ✖

Die richtige Antwort ist: 0,657609

Frage **8**
Richtig
Erreichte
Punkte 6,00
von 6,00
Frage
markieren

Which statements about Bidustering are correct?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. A common bidustering method is the variance maximization method.
- ☒ b. FABIA is a generative model. ✓
- ☐ c. FABIA defines bidclusters as inner products.
- ☒ d. In FABIA, the likelihood is analytically intractable and requires, e.g., the (variational) EM algorithm. ✓
- ☐ e. Row and column elements must belong to exactly one bidcluster or no bidcluster at all.
- ☒ f. Bidustering simultaneously clusters the rows and columns of a matrix. ✓

Die Antwort ist richtig.

Die richtigen Antworten sind: Bidustering simultaneously clusters the rows and columns of a matrix., FABIA is a generative model., In FABIA, the likelihood is analytically intractable and requires, e.g., the (variational) EM algorithm.